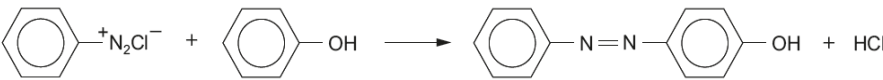


Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
10.	(a)			<table><tr><th>Stage</th><th>Reagent used</th><th>Structural formula of product</th></tr><tr><td>1</td><td>KBr / H₂SO₄ or SOCl₂ / PCl₅</td><td>CH₃CH₂CH₂Br CH₃CH₂CH₂Cl</td></tr><tr><td>2</td><td>KCN</td><td>CH₃CH₂CH₂CN</td></tr><tr><td>3</td><td>LiAlH₄</td><td></td></tr></table>	Stage	Reagent used	Structural formula of product	1	KBr / H ₂ SO ₄ or SOCl ₂ / PCl ₅	CH ₃ CH ₂ CH ₂ Br CH ₃ CH ₂ CH ₂ Cl	2	KCN	CH ₃ CH ₂ CH ₂ CN	3	LiAlH ₄		1	2		3		
				Stage	Reagent used	Structural formula of product																
				1	KBr / H ₂ SO ₄ or SOCl ₂ / PCl ₅	CH ₃ CH ₂ CH ₂ Br CH ₃ CH ₂ CH ₂ Cl																
				2	KCN	CH ₃ CH ₂ CH ₂ CN																
				3	LiAlH ₄																	
	(b)	(i)		sodium nitrate(III) / nitrite and acid e.g. HCl	1			1		1												
		(ii)		 accept other correct alternative equations		1		1														
	(c)			the statement is true as HCl will be removed as it is formed, moving the position of equilibrium to the right accept other valid reasons			1	1														